PREREQUISITES

Distributions NOT on formula sheet:

Name	pmf / pdf	Mean; Variance
Bin(m,p)	$\binom{m}{x}p^x(1-p)^{m-x}$	mp; $mp(1-p)$
Geo $x\geq1$	$p(1-p)^{x-1}$	$1/p$; $(1-p)/p^2$
Geo $x\geq0$	$p(1-p)^x$	$(1-p)/p$; $(1-p)/p^2$
NB(r,p)	$\binom{x+r-1}{x}p^r(1-p)^x$	$r(1-p)/p$; $r(1-p)/p^2$
Beta(a,b)	$\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}$	$\frac{a}{a+b}$; $\frac{ab}{(a+b)^2(a+b+1)}$

Variances the sheet omits: Exp(rate λ) $1/\lambda^2$; Gamma(a, rate b) a/b^2; InvGamma(a,b) $\frac{b^2}{(a-1)^2(a-2)}$ $(a>2)$; Poisson λ.
Gamma extras: $\mathbb{E}[1/X]=b/(a-1)$ $(a>1)$; $\mathbb{E}[\log X]=\psi(a)-\log b$, $\psi=\Gamma'/\Gamma$. Scale form: $x^{a-1}e^{-x/b}/(\Gamma(a)b^a)$ has mean ab. $\tau\sim$ Gamma(a,b) $\Leftrightarrow$ $1/\tau\sim$ InvGamma(a,b); if $\sigma^2\sim$ InvGamma(a,b), $\mathbb{E}[1/\sigma^2]=a/b$.
Bivariate Normal: $f(x)=\frac{1}{2\pi|\Sigma|^{1/2}}\exp\{-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\}$. $X_1|X_2=x_2\sim N(\mu_1+(x_2-\mu_2)\sigma_{12}/\sigma_2^2,\ \sigma_1^2-\sigma_{12}^2/\sigma_2^2)$.
Facts: $Z^2\sim\chi^2_1$; sum of k indep. $Z^2\sim\chi^2_k$ (mean k, var $2k$); $Z/\sqrt{U/r}\sim t_r$; $(U/m)/(V/n)\sim F_{m,n}$; $t_\nu^2\sim F_{1,\nu}$. $\sum$ Pois is Pois; $\sum_1^n$ Exp(λ) is Gamma(n,λ).
Critical values: z: 1.645 (90%), **1.96** (95%), 2.576 (99%). $\chi^2_{df}(.95)$: 1: **3.84**, 2: 5.99, 3: 7.81, 4: 9.49, 5: 11.07, 6: 12.59, 8: 15.51, 10: 18.31, 25: 37.65, 30: 43.77. $\chi^2_{df}(.99)$: 1: 6.63, 2: 9.21, 3: 11.34. $\Phi(1)=.84$, $\Phi(2)=.98$. $t_{60}(.975)=2.00$.

Confidence intervals: estimate ± critical value × SE.
Pivot: function of data and θ with fully known distribution.

Want	95% CI
μ, σ known	$\bar{x}\pm z\,\sigma/\sqrt{n}$
μ, σ unk.	$\bar{x}\pm t_{n-1}\,s/\sqrt{n}$
$\mu_X-\mu_Y$, eq. var	$\bar{x}-\bar{y}\pm t_{n+m-2}\,s_p\sqrt{1/n+1/m}$
p	$\hat{p}\pm z\sqrt{\hat{p}(1-\hat{p})/n}$
σ^2 (χ^2_{n-1} c)	$[(n-1)s^2/c_{.975},\ (n-1)s^2/c_{.025}]$
New X^* (PI)	$\bar{x}\pm t_{n-1}\,s\sqrt{1+1/n}$

Say: "if we repeated this, 95% of such intervals would contain the true value"; not "95% chance θ is here" (that is a credible interval). CI excludes 0 (1 for a ratio) $\Leftrightarrow$ reject H_0 at 5%.

Tests: H_0 no effect, H_1 effect. **Test statistic:** known distribution under H_0. Reject if statistic beyond the cutoff, or $p<\alpha$. **p-value:** P(this extreme or more | H_0), not P(H_0 true). **Type I:** reject true H_0 (α). **Type II:** keep false H_0 (β). **Power:** $1-\beta$. Write "reject" or "insufficient evidence", never "accept". χ^2, F right-tailed. Statistical ≠ practical significance.

Estimators: **Bias** $=\mathbb{E}\hat\theta-\theta$. **MSE** $=\mathrm{Var}\hat\theta+$ Bias^2. **Unbiased:** Bias 0. **Consistent:** $\hat\theta\to\theta$. **SE:** estimated sd of $\hat\theta$. **Precision:** 1/variance. $\bar{X}\approx N(\mu,\sigma^2/n)$ (CLT). $(n-1)S^2/\sigma^2\sim\chi^2_{n-1}$ for Normal data.

Model terms: **Nested:** A is B with parameters constrained (usually to 0). **Factor**, L levels: $L-1$ dummies vs a reference level; test it by one LRT on $L-1$ df, not its separate z values. `x1*x2` = `x1+x2+x1:x2`. `I(x^2)`: quadratic. Coefficients mean "holding other predictors fixed". Observational data show **association, not causation**. A predictor can flip sign when correlated with others: it is "not significant in the presence of the others".

LIKELIHOOD & BINOMIAL REGRESSION

Likelihood: independent Y_i with densities $f_i(\cdot;\theta)$ (may depend on covariates): $l(\theta)=\sum_i\log f_i(y_i;\theta)$. MLE $\hat\theta$ maximises l.
Score: $U(\theta)=\partial l/\partial\theta$. **Observed information:** $J(\theta)=-\partial^2 l/\partial\theta\,\partial\theta^\top=-H$, depends on y. **Fisher information:** $I(\theta)=\mathbb{E}J(\theta;Y)$, no y. This is the whole sample ($n\times$ one-obs info if iid). Bigger information = sharper curvature = smaller variance.
Asymptotics ($n\to\infty$, θ^* true): **Consistent:** $P(|\hat\theta-\theta^*|>\epsilon)\to0$. **Normal:** $\hat\theta\approx N(\theta^*,I(\theta^*)^{-1})$. **Efficient:** asymptotically unbiased with smallest variance $I(\theta^*)^{-1}$.
Regularity: l smooth (3rd derivatives exist, continuous, bounded expectation); support of Y free of θ; parameter space finite-dimensional, free of Y; θ^* not on boundary.
Q1 recipe: (a) l. (b) l', l''; $I=-\mathbb{E}[l'']$, replace $\sum x_i$ by $n\mathbb{E}X$ if data remain. (c) Solve $l'=0$, check $l''<0$, state $\hat\theta\approx N(\theta,1/I(\theta))$.
Marks: no data in l'': write "$J=I$, no expectation needed".
New parameter: the MLE plugs in (invariance), but I must be redone: one Bin: $I(p)=\frac{m}{p(1-p)}$, $I(\eta)=mp(1-p)$.
Course I is the whole sample: never divide by n again.

iid $X_i\sim$	MLE	$I(\theta)$
Exp(rate λ)	$1/\bar{x}$	n/λ^2
Exp(mean λ)	$\bar{x}$	n/λ^2
Poisson(λ)	$\bar{x}$	n/λ
$N(\mu_0,1/\tau)$	$n/\sum(x_i-\mu_0)^2$	$n/(2\tau^2)$
$N(\mu_0,\sigma^2)$	$\sum(x_i-\mu_0)^2/n$	$n/(2\sigma^4)$
$N(\mu,\sigma_0^2)$	$\bar{x}$	n/σ_0^2
Bin(m,p)	$\bar{x}/m$	$nm/(p(1-p))$
Geom $x\geq1$	$1/\bar{x}$	$n/(p^2(1-p))$
Geom $x\geq0$	$1/(1+\bar{x})$	$n/(p^2(1-p))$
Pois$(t_i\lambda)$	$\sum x_i/\sum t_i$	$\sum t_i/\lambda$
Bernoulli(p)	$\bar{x}$	$n/(p(1-p))$
$\theta x^{\theta-1}$ on (0,1)	$-n/\sum\log x_i$	n/θ^2
$\frac{x}{\theta}e^{-x^2/2\theta}$	$\sum x_i^2/2n$	n/θ^2
Gamma(a kn.,λ)	$a/\bar{x}$	na/λ^2
$\kappa(1+x)^{-\kappa-1}$	$n/\sum\log(1+x_i)$	n/κ^2
NegBin$(r,1-p)$, one y	$y/(y+r)$	$r/(p(1-p)^2)$

Wald CI for $t^\top\theta$: $\hat\theta\approx N(\theta^*,I(\hat\theta)^{-1})$, so $t^\top\hat\theta\pm z_\alpha\sqrt{t^\top I(\hat\theta)^{-1}t}$, $\Phi(z_\alpha)=1-\alpha/2$. Unit vector t: $\hat\theta_i\pm z_\alpha\sqrt{[I(\hat\theta)^{-1}]_{ii}}$. No I? Use $J(\hat\theta;y)$.
LR confidence region: $\{\theta: 2l(\hat\theta)-2l(\theta)\leq\chi^2_k(1-\alpha)\}$, $k=\dim\theta$. Better than the Normal approximation for small n. θ_0 is inside iff $2[l(\hat\theta)-l(\theta_0)]\leq\chi^2_k(.95)$. The Wald box ignores correlation: a point can sit inside both intervals yet outside the region.
Wald test: $H_0:\beta_j=0$. $z^*=\hat\beta_j/\mathrm{se}(\hat\beta_j)\sim N(0,1)$ under H_0. Reject if $|z^*|>1.96$.
Null not 0: $z=(\hat\beta-b_0)/$se; R's z column only tests 0.
LRT: Reduced R nested in Full F, k fewer parameters. $LR^*=-2[\log L(\hat\theta^R)-\log L(\hat\theta^F)]\sim\chi^2_k$ under R. If $LR^*>\chi^2_k(1-\alpha)$ reject R, pick F.
Wald vs LRT: asymptotically equivalent ($z^{*2}\approx\chi^2_1$), but the χ^2 approximation to the LR is generally better, so prefer LRT.

Binomial regression: $Y_i\sim\mathrm{bin}(m_i,p_i)$ independent, m_i known (binary: $m_i=1$), $g(p_i)=\eta_i=x_i^\top\beta$. Linear $p=\beta_0+\beta_1t$ fails because $0\leq p\leq1$ is needed.
Logit: $\eta=\log\frac{p}{1-p}$, $p=\frac{e^\eta}{1+e^\eta}=\frac{1}{1+e^{-\eta}}$. **Probit:** $\eta=\Phi^{-1}(p)$, $p=\Phi(\eta)$. **Cloglog:** $\eta=\log(-\log(1-p))$, $p=1-e^{-e^\eta}$.
Logistic curve $\eta=\beta_0+\beta_1t$: $p=\frac{1}{2}$ at $t=-\beta_0/\beta_1$ (location), slope there $\beta_1/4$ (steepness).

$$l(\beta)=c+\sum_i\left[y_i\log\frac{p_i}{1-p_i}+m_i\log(1-p_i)\right]$$

Logit: $l=c+\sum_i[y_i\eta_i-m_i\log(1+e^{\eta_i})]$. No closed form: use `optim` or IWLS.
Logit information: $J=I=X^\top WX$, $W=\mathrm{diag}(m_ip_i(1-p_i))$ (no y_i left). One predictor: entries $\sum m_ip_iq_i$, $\sum m_ix_ip_iq_i$, $\sum m_ix_i^2p_iq_i$, $q_i=1-p_i$.
Probit: $I=X^\top WX$, $W_{ii}=m_i\varphi(\eta_i)^2/[\Phi(\eta_i)(1-\Phi(\eta_i))]$.
Predict at x_0: $\hat\eta_0=x_0^\top\hat\beta$, $\hat{p}_0=g^{-1}(\hat\eta_0)$.
CI for p: p is not linear in β, so (1) CI for η: $\hat\eta_0\pm1.96\sqrt{x_0^\top Vx_0}$, $V=I(\hat\beta)^{-1}$; one predictor $x_0^\top Vx_0=V_{11}+2x_0V_{12}+x_0^2V_{22}$. (2) Apply g^{-1} to both ends (g monotone).
Two predictors, $x_0=(1,a,b)$: $x_0^\top Vx_0=V_{00}+a^2V_{aa}+b^2V_{bb}+2aV_{0a}+2bV_{0b}+2abV_{ab}$. Keep every digit of `vcov`: terms nearly cancel; drop the covariances and se is far off. Smallest near the centre of the xs, wide when extrapolating.
Odds: $\log o=\eta$. Change x_j by Δ: $OR=e^{\Delta\beta_j}$, CI $\exp\{\Delta(\hat\beta_j\pm1.96\,\mathrm{se})\}$ (swap ends if $\Delta<0$). Several x change by vector d: $\log OR=d^\top\hat\beta$, se $=\sqrt{d^\top Vd}$.
OR between two non-baseline levels: $\log OR=\hat\beta_a-\hat\beta_b$, se $=\sqrt{V_{aa}+V_{bb}-2V_{ab}}$.
Classify: $\hat{p}\geq$ cutoff; trade off false positives and negatives by cost (screening: fewer false negatives).
Score (logit): $\sum_i(y_i-m_ip_i)x_{ij}=0$: observed minus fitted.
Links: same predictors with different links are not nested: no LRT, use AIC. They agree mid-range, differ in the tails. Cloglog leaves 0 slowly, nears 1 fast; natural if $p=1-e^{-\lambda}$ (at least one Poisson event). Probit, cloglog: e^β is not an OR; y stays in the Hessian, so $J\neq I$.
Case-control: logit slopes are still valid log OR; only the intercept shifts, by $\log(\pi_1/\pi_0)$, so no population risks. Not true for probit.
Sparse data (Hauck-Donner): se too big, z too small, a real effect missed: trust ΔD.

GLMs: DEVIANCE, IWLS, DIAGNOSTICS

Deviance: **Saturated S:** one parameter per observation, $\hat\mu_i=y_i$. **Null:** intercept only. **Scaled deviance:** $D^A/\phi=-2\log[L(\hat\beta_A)/L(S)]=2(l_S-l_A)$. **Deviance** is $D^A=\sum d_i$. Binomial and Poisson: $\phi=1$, same thing. $D_{null}-D_{res}$ = improvement from predictors. **Why (Poisson):** $l_S=\sum[y_i\log y_i-y_i-\log y_i!]$, $\hat{l}=\sum[y_i\log\hat\mu_i-\hat\mu_i-\log y_i!]$; double the gap.
$y=0$: Poisson $d_i=2\hat\mu_i$; binomial $d_i=-2m_i\log(1-\hat{p}_i)$.

Model	d_i ($0\log0=0$)
Normal	$(y_i-\hat\mu_i)^2$ (= SSE in LM)
Poisson	$2[y_i\log(y_i/\hat\mu_i)-(y_i-\hat\mu_i)]$
Binomial	$2[y_i\log\frac{y_i}{\hat\mu_i}+(m_i-y_i)\log\frac{m_i-y_i}{m_i-\hat\mu_i}]$
Gamma	$2[-\log(y_i/\hat\mu_i)+(y_i-\hat\mu_i)/\hat\mu_i]$

Adequacy (goodness of fit), $\phi=1$ only: H_0: fitted model adequate. If m_ip_i, $m_i(1-p_i)\geq5$ (binomial) or $\lambda_i\geq5$ (Poisson) then $D\approx\chi^2_{n-p}$, p including β_0. Reject if $D>\chi^2_{n-p}(.95)$; rough check $D\approx df$. Saturated model has n parameters (not fixed), so ML theory does not apply: D need not be χ^2. Small m_i or counts (binary $m_i=1$ especially): no adequacy test, ΔD still fine. Normal, Gamma, IG: ϕ estimated, cannot test adequacy. D far above the cutoff: misfit is real even if a few counts are small. 0/1 logit: D depends only on $\hat{p}_i$, so says nothing about fit.
Lack of fit may be overdispersion, wrong mean or missing interaction: check residuals vs $\hat\eta$ first.
Nested: A in B, s extra parameters. $(D^A-D^B)/\phi=-2[l_A-l_B]$, l_S cancels. $\phi=1$: $D^A-D^B\approx\chi^2_s$. ϕ unknown: F test (see Overdispersion). Subtract residual deviances, not null deviances. $D_{null}-D_{res}$ on $p-1$ df tests all predictors at once. **Numeric score vs factor** (L levels), or a group-level covariate vs a group factor: nested, $s=L-2$.
AIC (non-nested too): $AIC=2p-2\hat{l}$, smaller is better, same data only, not a test. B has s more parameters: $AIC_B-AIC_A=2s-(D_A-D_B)/\phi$ (exact for $\phi=1$). An extra parameter pays only if it cuts D by more than 2.
From R output: $p=n-$residual df; $n=$ null df $+1$. $\phi=1$: $\hat{l}=p-AIC/2$, $l_S=\hat{l}+D/2$ (same l_S for every model on those data). Gamma, Normal: R's AIC counts ϕ, so $\hat{l}=(p+1)-AIC/2$. Check: two models on the same data give the same l_S. Continuous data: $\hat{l}>0$, AIC <0 are fine.

Exponential family: $f(y;\theta,\phi)=\exp\left[\frac{y\theta-b(\theta)}{a(\phi)}+c(y,\phi)\right]$. θ canonical (location), ϕ dispersion (scale). Form not unique. Convention $a(\phi)=\phi$.
$\mathbb{E}Y=b'(\theta)$, $\mathrm{Var}Y=b''(\theta)a(\phi)=v(\mu)a(\phi)$. **Variance function** $v(\mu)=b''((b')^{-1}(\mu))$: how spread changes with mean. **Canonical link** $g=(b')^{-1}$, $g(\mu)=\theta$. Canonical: $g'(\mu)=1/v(\mu)$.
Proof: $\int f\,dy=1$. Differentiate in θ: $\mathbb{E}[(Y-b')/a]=0$, so $\mathbb{E}Y=b'$. Again: $\mathbb{E}[(Y-b')^2/a^2]-b''/a=0$, so $\mathrm{Var}Y=b''a$.
Recipe: write $\log f$; find $y\times$(something) $=\theta/a$; the term with no y gives $-b/a$; leftovers c. Write b in θ only. Check $b'(\theta)=\mu$. State all of θ,ϕ,a,b,c and write "support free of θ". Not exp family if the support depends on θ, or the parameter multiplies y^2, $\log y$ or e^y rather than y.

- **Normal:** $\theta=\mu$, $\phi=\sigma^2$, $b=\theta^2/2$, $c=-\frac{1}{2}(\frac{y^2}{\phi}+\log2\pi\phi)$; $g=\mu$, $v=1$.
- **Poisson:** $\theta=\log\lambda$, $\phi=1$, $b=e^\theta$, $c=-\log y!$; $g=\log\mu$, $v=\mu$.
- **Binomial:** $\theta=\log\frac{p}{1-p}$, $\phi=1$, $b=m\log(1+e^\theta)$, $c=\log\binom{m}{y}$; $g=\log\frac{\mu}{m-\mu}$, $v=\mu(1-\mu/m)$.
- **Gamma**(ν,λ): $\theta=-\lambda/\nu=-1/\mu$, $\phi=1/\nu$, $b=-\log(-\theta)$, $c=(\frac{1}{\phi}-1)\log y-\frac{\log\phi}{\phi}-\log\Gamma(\frac{1}{\phi})$; $g=-1/\mu$ (R: $1/\mu$), $v=\mu^2$, $\mathrm{Var}Y=\phi\mu^2$.
- **Exponential**(λ): $\theta=-\lambda$, $\phi=1$, $b=-\log(-\theta)$, $c=0$; $g=-1/\mu$, $v=\mu^2$. $\mu=-1/\eta$ needs $\eta<0$, so canonical link can fail: use log. ϕ known: χ^2 tests.
- **Inverse Gaussian:** $\theta=-\frac{1}{2\mu^2}$, $\phi=1/\lambda$, $b=-\sqrt{-2\theta}$, $c=-\frac{1}{2}[\log(2\pi\phi y^3)+\frac{1}{\phi y}]$; $g=-\frac{1}{2\mu^2}$ (R: $1/\mu^2$), $v=\mu^3$.
- **Weibull,** k known: $Y=X^k\sim$ Exp(rate λ^k).
- **Geometric,** $y\geq0$: $\theta=\log(1-p)$, $\phi=1$, $b=-\log(1-e^\theta)$, $c=0$; $v=\mu+\mu^2$.
- **NegBin,** r known: $\theta=\log\frac{\mu}{\mu+r}$, $\phi=1$, $b=-r\log(1-e^\theta)$, $c=\log\frac{\Gamma(y+r)}{\Gamma(r)y!}$; $v=\mu+\mu^2/r$. r unknown: not exp family.
- **Proportion** $Y=X/m$: θ logit, $b=\log(1+e^\theta)$, $a(\phi)=1/m$, $v=\mu(1-\mu)$ (R's scale). **Mean of k Poissons:** $b=e^\theta$, $a(\phi)=1/k$, $\mathrm{Var}Y=\mu/k$; ϕ known from the design.

GLM: Y from exponential family and $\mu=\mathbb{E}Y=g^{-1}(x^\top\beta)$; g monotone differentiable link. We model location $\eta=x^\top\beta$; scale sorts itself out. Canonical link: $\theta_i=x_i^\top\beta$.
Fitted mean: log link e^η; logit $me^\eta/(1+e^\eta)$; identity η; inverse $1/\eta$.
Newton-Raphson: $x_{n+1}=x_n-f'(x_n)/f''(x_n)$; $\theta^{n+1}=\theta^n-H(\theta^n)^{-1}U(\theta^n)$. **Fisher scoring:** replace J by I (positive definite): $\theta^{n+1}=\theta^n+I(\theta^n)^{-1}U(\theta^n)$. Equal under a canonical link.
Score: $U_j=\sum_i\frac{(y_i-\mu_i)x_{ij}}{a(\phi)v(\mu_i)g'(\mu_i)}$. Canonical link: $X^\top(y-\hat\mu)=0$, so $\sum\hat\mu_i=\sum y_i$ with an intercept.
Canonical link: $W=v(\mu)$, $J=I$, NR = Fisher scoring = IWLS, $X^\top y$ sufficient, and fitted totals match the data: group totals, $\sum x_{ij}\hat\mu_i=\sum x_{ij}y_i$. Non-canonical: J contains y; same $\hat\beta$.
WLS: $Y=X\beta+\varepsilon$, $\varepsilon\sim N(0,\Sigma)$: $\hat\beta=(X^\top\Sigma^{-1}X)^{-1}X^\top\Sigma^{-1}y$. **Idea:** $g(Y_i)\approx Z_i=g(\mu_i)+(Y_i-\mu_i)g'(\mu_i)$ with Var $=g'(\mu_i)^2v(\mu_i)a(\phi)$; depends on β, so iterate.
IWLS: (1) $\hat\mu^0=y$. (2) $z_i=g(\hat\mu_i)+(y_i-\hat\mu_i)g'(\hat\mu_i)$, $W_{ii}=1/[g'(\hat\mu_i)^2v(\hat\mu_i)]$. (3) $\hat\beta=(X^\top WX)^{-1}X^\top Wz$, $\hat\mu_i=g^{-1}(x_i^\top\hat\beta)$. (4) Repeat until $\hat\beta$ stable. $a(\phi)$ cancels. Same as Fisher scoring. Needs only g, v.
One step by hand: $\hat\mu^0=y$, so $z_i=g(y_i)$, $W_{ii}=1/[g'(y_i)^2v(y_i)]$ (Poisson log: $\log y$, y). $X^\top WX=\begin{pmatrix}\sum W & \sum Wx\\ \sum Wx & \sum Wx^2\end{pmatrix}$, $X^\top Wz=\begin{pmatrix}\sum Wz\\ \sum Wxz\end{pmatrix}$. $y=0$ breaks the start: R uses $y+.1$ (Poisson), $(y+.5)/(m+1)$ (binomial).

Family, link	z_i	W_{ii}
Poisson, log	$\eta+(y-\mu)/\mu$	μ
Binom, logit	$\eta+\frac{(y-\mu)m}{\mu(m-\mu)}$	$\mu(m-\mu)/m$
Gamma, log	$\eta+(y-\mu)/\mu$	1
Poisson, $\sqrt{\ }$	$\eta/2+y/(2\eta)$	4
Binom, cloglog	$\eta+\frac{y-\mu}{m(1-p)L}$	$m(1-p)L^2/p$
Gamma, $1/\mu$	$\eta-(y-\mu)/\mu^2$	μ^2
Gamma, identity	y	$1/\mu^2$
IG, $1/\mu$	$\eta-(y-\mu)/\mu^2$	μ

Cloglog: $L=e^\eta=-\log(1-p)$.
Variance: $\mathrm{Var}\hat\beta=(X^\top\hat\Sigma^{-1}X)^{-1}$, $\hat\Sigma_{ii}=g'(\hat\mu_i)^2v(\hat\mu_i)a(\hat\phi)$. Same as $a(\hat\phi)(X^\top\hat{W}X)^{-1}$: $a(\phi)$ does not cancel here. Gamma log: $\hat\phi(X^\top X)^{-1}$. ϕ known but R took 1: SE$\times\sqrt{a(\phi)}$.
Estimate ϕ: $\hat\phi=X^2/(n-p)$, Pearson $X^2=\sum_i(y_i-\hat\mu_i)^2/v(\hat\mu_i)$. Consistent. Not $D/(n-p)$. If $\phi=1$ and the model is adequate, $X^2\approx\chi^2_{n-p}$.
Rate model: $Y_i\sim\mathrm{Pois}(\lambda_i)$, $\lambda_i\propto t_i$: $\log\lambda_i=\log t_i+x_i^\top\beta$. `offset(log(t))` fixes the coefficient of $\log t$ at 1 (uses no df). e^β = rate ratio; $\hat\lambda=te^{x^\top\hat\beta}$. **Check it:** free coefficient of $\log t$, $z=(\hat\beta-1)/$se, or ΔD on 1 df. Forget $\log t$ in $\hat\eta$ and you get the rate, not the count.
Gamma GLM: `Gamma(link="log")` etc. ϕ estimated, so t and F. sd$/\mu=\sqrt{\phi}$. Pick link by plotting $g(y)$ vs x.
Intercept only: $\hat\mu=\bar{y}$, $\hat\beta_0=g(\bar{y})$; Poisson log se $=1/\sqrt{\sum y_i}$. **Quadratic:** turning point $x=-\hat\beta_1/(2\hat\beta_2)$. **Log link with** $\log x$: $\mu=e^{\beta_0}x_1^{\beta_1}x_2^{\beta_2}$; test a theory value by $(\hat\beta-b_0)/$se. **Identity link:** CI directly on the response scale. **Inverse link:** larger η means smaller μ.

Residuals: **Response:** $y_i-\hat\mu_i$; not homoskedastic unless v constant. **Pearson:** $r_P(i)=(y_i-\hat\mu_i)/\sqrt{v(\hat\mu_i)}$, $\sum r_P^2=X^2$. **Deviance:** $r_D(i)=\mathrm{sign}(y_i-\hat\mu_i)\sqrt{d_i}$, $\sum r_D^2=D$ (R default). Binomial $v=m\hat{p}\hat{q}$. $y=0$: $r_D=-\sqrt{2\hat\mu}$, a smooth curve on the plot. `type="response"` on `cbind` gives $y/m-\hat{p}$.
Leverage: $H=X(X^\top\hat\Sigma^{-1}X)^{-1}X^\top\hat\Sigma^{-1}$, H_{ii} measures potential influence (unusual x; also depends on IWLS weight), no ϕ. $\sum H_{ii}=p$: rule of thumb, flag $>2p/n$. Large leverage need not mean actual influence. By hand: $H_{ii}=w_ix_i^\top\widehat{\mathrm{Var}}(\hat\beta)x_i$ ($\phi=1$), w_i the IWLS weight: $m\hat{p}\hat{q}$ (logit), $\hat\mu$ (Poisson log), 1 (Gamma log: H depends on x only). Mean leverage p/n. Tiny $\hat\mu$: small H even at extreme x.
Studentised: $\mathrm{Var}(Z-\hat{Z})=(I-H)\Sigma$, so $r_{SP}=r_P/\sqrt{(1-H_{ii})\hat\phi}$, $r_{SD}=r_D/\sqrt{(1-H_{ii})\hat\phi}$. Unit variance after adjusting for leverage. ±: above/below fit.
Jack-knife: change in $\hat\mu_i$ when y_i removed (outliers): $\mathrm{sign}(y_i-\hat\mu_i)\sqrt{(1-H_{ii})r_{SD}^2+H_{ii}r_{SP}^2}$. $\phi=1$: sign $\cdot\sqrt{r_D^2+H_{ii}r_P^2/(1-H_{ii})}$.
Cook: change in the whole fit when i removed: $(\hat\beta_{(i)}-\hat\beta)^\top X^\top\hat\Sigma^{-1}X(\hat\beta_{(i)}-\hat\beta)/p\approx r_{SP}^2H_{ii}/[p(1-H_{ii})]$. Same as $r_P^2H_{ii}/[p\hat\phi(1-H_{ii})^2]$. Flag near 1 or a clear jump. Gamma: keep $\hat\phi$.
Half-normal plot: ordered |values| vs $\Phi^{-1}((n+i)/(2n+1))$. Smooth = fine; jump or kink in tail = problem. Need not be straight.
r_D vs $\hat\eta$: no trend, even spread is good. Trend or curve: wrong mean. Fan: wrong variance. **Banding:** discrete counts, not a problem; use a smoother. 0/1 data: two curves, nothing to read.
Linearity: plot $g(y)$ or z vs each x_j: want straight. Fix by transforming x, not y (breaks distribution of Y).
Influential point: refit without it, compare $\hat\beta$, D, $\hat\phi$. Typically keep unless a recording error.
Classify: **outlier:** big jack-knife, ordinary H, coefficients barely move. **High leverage only:** big H, tiny residual, small Cook. **Influential:** big Cook, coefficients move when dropped (needs both leverage and residual).

OVERDISPERSION & QUASI-LIKELIHOOD

Overdispersion: actual $\mathrm{Var}Y_i$ larger than the model says.
Causes: binomial: dependent trials or p changing trial to trial. Poisson: dependent events or non-constant rate. Random rate: $\mathrm{Var}Y=\mathbb{E}\lambda+\mathrm{Var}\lambda>\mathbb{E}\lambda$. Clusters of size k, $\mathrm{Var}p_i=\tau^2p(1-p)$: $\mathrm{Var}Y=(1+(k-1)\tau^2)mp(1-p)$.
Detect: $D\gg df$, GOF p tiny, $\hat\phi\gg1$ (no universal cutoff). Rule out wrong mean and outliers first. D/df is close to $\hat\phi$, but quote Pearson. A free mean for every group and D still big: must be overdispersion. 0/1 data: cannot be detected.
If ignored: $\hat\beta$ unaffected, but SEs too small, z too big, p-values too small, CIs too narrow: spurious significant predictors, too big a model. True se $\approx\sqrt{\hat\phi}$ se, $z\to z/\sqrt{\hat\phi}$, $\Delta D\to\Delta D/\hat\phi$. $\hat\phi>1$ only ever makes terms less significant; $\hat\phi<1$: SEs too large.
Fixes: quasi-likelihood; different model (Poisson with Gamma rate gives NegBin, $\mathrm{Var}=\mu+\mu^2/\nu$); Bayes: random effect in η.
Quasi-likelihood: specify only $g(\mu_i)=x_i^\top\beta$, $\mathbb{E}Y_i=\mu_i$, $\mathrm{Var}Y_i=\phi v(\mu_i)$; no distribution. Score $u_i=\frac{Y_i-\mu_i}{\phi v(\mu_i)}$: $\mathbb{E}u=0$, $\mathrm{Var}u=\frac{1}{\phi v}$, $-\mathbb{E}\frac{\partial u}{\partial\mu}=\frac{1}{\phi v}=\mathbb{E}u^2$, same as $\partial\log L/\partial\mu$.

$$Q_i=\int_{y_i}^{\mu_i}\frac{y_i-t}{\phi v(t)}dt,\quad Q=\sum_i Q_i$$

$\partial Q_i/\partial\mu_i=u_i$. Q need not be a real log-density. $\hat\beta$ is consistent and asymptotically Normal as for MLE, but less efficient: use the distribution if known.
Quasi-deviance: saturated $Q=0$, so scaled $D_Q/\phi=-2\sum Q_i$, $D_Q=-2\sum_i\int_{y_i}^{\mu_i}\frac{y_i-t}{v(t)}dt$.
Quasi-Poisson: $v=\mu$, $Q_i=\frac{1}{\phi}[y_i\log\mu_i-\mu_i]$ + const. **Quasi-binomial:** $v=\mu(m-\mu)/m$, $Q_i=\frac{1}{\phi}[y_i\log\frac{\mu_i}{m_i-\mu_i}+m_i\log(m_i-\mu_i)]$ + const. ϕ cancels in IWLS, so same $\hat\beta$, $\hat\mu$, D as Poisson/binomial.
Numeric Q_i (definite integral, ≤ 0): $v=\mu$: $\phi Q_i=y\log\frac{\mu}{y}-(\mu-y)$. $v=\mu(m-\mu)/m$: $y\log\frac{\mu}{y}+(m-y)\log\frac{m-\mu}{m-y}$. $v=\mu^2$: $1-\frac{y}{\mu}+\log\frac{y}{\mu}$. $v=\mu^3$: $-\frac{(y-\mu)^2}{2y\mu^2}$.
Other v: $v=1$: $\phi Q_i=-\frac{1}{2}(y-\mu)^2$, D_Q = RSS, $\hat\phi=\hat\sigma^2$. D_Q for $v=\mu$, $\mu(m-\mu)/m$, μ^2, μ^3 is the Poisson, binomial, Gamma, IG deviance. $Q_i\leq0$: integrand opposes the direction of integration. **Why Pearson, not max Q:** $Q=\frac{1}{\phi}\times(\leq0)$ rises to 0 as $\phi\to\infty$. **Scaled:** $D_Q/\hat\phi=-2\sum Q_i\approx n-p$ by construction, so no fit test.
SE: $\mathrm{se}_Q=\sqrt{\hat\phi}\,\mathrm{se}_P$, $\hat\phi$ of that same model; reverse: $\mathrm{se}_P=\mathrm{se}_Q/\sqrt{\hat\phi}$. Estimate never changes. $t=z/\sqrt{\hat\phi}$, $z=t\sqrt{\hat\phi}$.
One point: $\mathrm{Var}Y_i=\hat\phi m\hat{p}\hat{q}$.
Coefficient: $t^*=\hat\beta/\mathrm{se}_Q\approx t_{n-p}$; CI $\hat\beta\pm t_{n-p}(.975)\mathrm{se}_Q$ (1.96 if no t given), exp for ratio. Prediction is unchanged, only its CI widens: `predict` on the quasi fit includes $\hat\phi$ (`residual.scale` $=\sqrt{\hat\phi}$); on the Poisson fit multiply se.fit by $\sqrt{\hat\phi}$.
Nested F: $F=\frac{(D_A-D_B)/s}{\hat\phi}\approx F_{s,n-p_B}$, $\hat\phi$ from the BIG model B. No adequacy test, no χ^2, AIC is NA (no likelihood), `step`/`drop1` do not work. Exact for Normal (LM F test), approximate otherwise. Big model's $\hat\phi$ not printed: approx. $D_B/(n-p_B)$. X^2 can fall while $\hat\phi$ rises: df fell, the extra terms explain nothing.
New data type: choose link and v from mean-variance (e.g. proportions: logit, $\mu(1-\mu)$), fit IWLS, $\hat\phi$ from Pearson. $\hat\phi<1$ (underdispersion) is allowed. iid mean λ, var $\phi\lambda$: $\hat\lambda=\bar{y}$, $\hat\phi=\frac{\sum(y_i-\bar{y})^2}{\bar{y}(n-1)}$. Non-integer fractions work in `quasibinomial`. Lognormal: $v=\mu^2$, $\phi=e^{\sigma^2}-1$.

READING R OUTPUT

`summary(glm)` line by line:

- **Deviance Residuals** (old R): 5-number summary of r_D; want median near 0, roughly symmetric.
- **Estimate** $\hat\beta_j$: change in η per unit of x_j, others fixed. Row `schoolB` = level B vs baseline; `a:b` = interaction; `(Intercept)` = η at $x=0$, baseline levels. **Std. Error:** $\sqrt{V_{jj}}=\sqrt{[I(\hat\beta)^{-1}]_{jj}}$.
- **z value** (t if ϕ estimated) = Est/SE: tests $\beta_j=0$ with every other term kept in. **Pr(>|z|):** two-sided p. Stars `***` $<.001$, `**` $<.01$, `*` $<.05$, `.` $<.1$.
- **Dispersion line:** "taken to be 1" is an assumption, not an estimate. Other value: $\hat\phi=X^2/(n-p)$.
- **Null deviance** on $n-1$: intercept-only fit, same for every model on those data. **Residual deviance** on $n-p$: your model.
- **AIC:** $2p-2\hat{l}$; NA for quasi. **Fisher Scoring iterations:** IWLS steps. **"k observations deleted":** n is smaller than the data.

Recover anything hidden: $n=$ null df $+1$ (with intercept). $p=n-$ res df. Non-intercept terms = null df − res df. SE = Est/z; Est $=z$SE. $\hat{l}=p-AIC/2$ (Gamma $p+1$); $l_S=\hat{l}+D/2$; $\hat{l}_B=\hat{l}_A+\Delta D/2$. $s=$ res df$_A-$ res df$_B$. $\Delta D=D_A-D_B$. $AIC_B-AIC_A=2s-\Delta D/\hat\phi$. $\hat\phi=X^2/(n-p)=(\mathrm{se}_Q/\mathrm{se}_P)^2=(z/t)^2$; $X^2=\hat\phi(n-p)$. $F=(\Delta D/s)/\hat\phi$, so $\hat\phi=(\Delta D/s)/F$. $z^2\approx\Delta D$ and $t^2\approx F$ on 1 df. Logit $\hat{p}=1/(1+e^{-\hat\eta})$; log link $\hat\mu=e^{\hat\eta}$. $|z|>1.96$: $p<.05$; $|z|>2.58$: $p<.01$.

`anova(A,B,test="Chisq")`, two models: row 1 is the smaller model A, row 2 the bigger B.

- **Resid. Df:** $n-p_A$, $n-p_B$. **Resid. Dev:** D_A, D_B.
- **Df** (row 2): s. **Deviance:** $\Delta D=D_A-D_B\geq0$.
- **Pr(>Chi):** $P(\chi^2_s>\Delta D)$. Small: reject A, keep B. Large: keep A.
- `test="F"` adds **F** $=(\Delta D/s)/\hat\phi_B$ and **Pr(>F)** on $(s,n-p_B)$.

`anova(m,test="Chisq")`, one model: "terms added sequentially (first to last)".

- **NULL row:** Resid. Df $n-1$, Resid. Dev D_{null}.
- **Term rows:** that term added after those above it. Df its df; Deviance the drop in D; Resid. Df, Resid. Dev running totals.
- Order matters: only the last row tests a term given all the others.
- **Checks:** Df column sums to $p-1$; Deviance column to $D_{null}-D_{res}$; last Resid. Dev is the summary's.
- Read upward: interaction first. If `A:B` is significant keep `A` and `B`, whatever their rows say.

`drop1(m,test="Chisq")`: `<none>` is m; each row drops only that term, so order is irrelevant. Deviance = D after dropping; LRT = rise in D; 1-df term: LRT $\approx z^2$. `a*b`: only `a:b` shown.
`step(m)`: `Start: AIC=` is the current model. Rows sorted by AIC, lowest first: `- x` drops x, `<none>` changes nothing. Top row wins; stops when `<none>` is on top. Last block's formula is the final model. Fails on quasi (AIC NA).

Which test goes with which family:

	Poisson, binomial	Gamma, IG	Quasi
ϕ	1, fixed	$\hat\phi$ Pearson	$\hat\phi$ Pearson
Coef	z	t_{n-p}	t_{n-p}
Nested	ΔD vs χ^2_s	$F_{s,n-p_B}$	$F_{s,n-p_B}$
Fit	D vs χ^2_{n-p}, fitted ≥5	none	none
$\hat{l}$	$p-AIC/2$	$p+1-AIC/2$	AIC NA

Degrees of freedom: **null** $n-1$. **Residual, GOF χ^2, Wald t, $\hat\phi$:** $n-p$, p counting the intercept. **All terms:** $p-1$. **LRT χ^2:** s = difference in residual df. **Quasi, Gamma F:** $(s,n-p_B)$. **Numeric term** or `I(x^2)`: 1. **Offset:** 0. **Factor:** $L-1$. **Two factors** `A:B`: $(L_A-1)(L_B-1)$. **Factor by numeric** `A:x`: $L-1$. `A*B`: main effects plus interaction. **Numeric to factor:** $L-2$. **LR region:** $\dim\theta$. **Saturated:** 0 residual df. **Log-linear table:** cells − parameters; `A+B+C` = mutual independence, `A*B` = interaction. **Mixture:** $K-1$ + component parameters.

Interpretation lines: **Log link:** "Holding other predictors fixed, a 1-unit rise in x multiplies the expected count by $e^{\hat\beta}$, a $100(e^{\hat\beta}-1)$% change." **Logit:** same for the odds, not the probability. **Identity:** adds $\hat\beta$. **Inverse:** $1/\mu$ rises by $\hat\beta$, so $\hat\beta>0$ lowers μ. **Factor:** "relative to the reference level". $\log x$ **with log link:** doubling x multiplies μ by $2^{\hat\beta}$. **Interaction:** the effect of one predictor depends on the other; a cell vs baseline is exp(its main effects + interaction); add the intercept for the cell mean. **Intercept:** e^{β_0} = baseline mean, rate or odds at $x=0$. **Δ units:** $e^{\Delta\hat\beta}$, not $\Delta e^{\hat\beta}$. CI: exponentiate the ends, never the se. Observational: "associated with", not "causes".

Hard subquestions (R output):

- **Which model, 2 quantitative reasons:** smaller AIC, and $\Delta D>\chi^2_s(.95)$ rejects the smaller one.
- **2 reasons overdispersed:** $D\gg df$ and $\hat\phi\gg1$.
- **Why F, t not χ^2, z:** ϕ unknown, estimated by $\hat\phi$.
- **Poisson vs quasi-Poisson:** Poisson fixes $\mathrm{Var}Y=\mu$. Quasi keeps the same mean model, sets $\mathrm{Var}Y=\phi\mu$ with ϕ estimated from Pearson and assumes no distribution. Same $\hat\beta$; SEs $\times\sqrt{\hat\phi}$; tests become t, F; no AIC.
- **Interaction tested twice:** z from Fisher information $\sim N(0,1)$, and $\Delta D\sim\chi^2_1$. Prefer ΔD.
- **Linear assumption evidence:** $\log(y/t)$ vs x straight; interaction plot parallel lines = no interaction.
- **Outlier:** clear point in half-normal Cook/jack-knife plot. Refit, re-check $\hat\phi$: overdispersion typically remains.
- **GOF valid?** Check first: fitted counts $\hat\mu$ (binomial $m\hat{p}$, $m\hat{q}$) all ≥ 5; never for 0/1.
- **χ^2 says keep, F says drop:** believe F, $\phi=1$ is false: prefer the smaller model.
- **Which $\hat\phi$?** SE: the model holding that coefficient. F: the bigger model.
- **Factor with no significant dummy:** can still matter: one LRT on $L-1$ df.
- **$z^2\neq\Delta D$, $t^2\neq F$:** curvature at $\hat\beta$ vs the actual drop; close, not equal. Trust ΔD.

Maid Mast Codes PT1: GLM

`glm(cbind(y,m-y)~x, family=binomial, d)`: grouped binomial. `family=poisson`, `quasipoisson`, `quasibinomial`, `Gamma(link="log")`. `subset=-25` drops row 25; `update(m, subset=-7)` refits without row 7.

- `anova(A,B,dispersion=phi,test="F")` on binomial fits: same F as quasi, but p uses $F_{s,\infty}$. `test="F"` on plain Poisson/binomial: $F=\Delta D/s$, ignores overdispersion (R warns).
- `sum(residuals(m,type="pearson")^2)/m$df.residual` $=\hat\phi$.
- `pchisq(D,df,lower.tail=FALSE)` $=P(\chi^2_{df}>D)$. `qchisq(.95,df)`, `qf(.95,s,n-p)`, `qt(.975,n-p)`: cutoffs. `predict(m,newdata=list(x=1),type="response")`: $\hat\mu$ (`cbind` binomial: $\hat{p}$, multiply by m; same for `fitted`; offset: `newdata` needs t). `type="link",se.fit=TRUE`: $\hat\eta$, se. `rstudent` jack-knife (= the formula if $\phi=1$); `influence(m)$hat`; `cooks.distance`; `halfnorm`.
- `sum(m*fitted(f)<5)`: rows with $m\hat{p}<5$; check $m\hat{q}$ too. `logLik(m)`: $\hat{l}$ with df p (Gamma $p+1$, quasi NA). `cov2cor(vcov(m))`: correlations. `binomial()$variance(mu)`: v on the proportion scale.
- `optim(start,logL,control=list(fnscale=-1))` maximises l; start values matter.
- `summary(m,dispersion=phi)` rescales SEs ($z\div\sqrt{\phi}$, uses Normal). Old papers: `drop1(m,scale=phi,test="F")`, its F uses $D/(n-p)$, not Pearson.
- `vcov(m)` $=V$ (quasi includes $\hat\phi$); `rstandard(m)` $=r_{SD}$; `rstandard(m,type="pearson")` $=r_{SP}$; `hatvalues(m)` $=H_{ii}$.
- `z<-predict(m)+(y-mu)/mu`: working response, log link. `w<-mu`: weight. `dnorm`, `rnorm` take sd; `rgamma(n,shape,rate)`.

PROBABILITY TOOLKIT

Probability toolkit: **Total prob:** $P(A)=\sum_k P(A|B_k)P(B_k)$. **Bayes:** $P(B_k|A)=P(A|B_k)P(B_k)/\sum_j P(A|B_j)P(B_j)$. **Joint:** $p(x,z)=p(x|z)p(z)$. **Marginal:** $p(x)=\int p(x,z)\,dz$ (or $\sum_z$). **Conditional indep.** given θ: $p(\tilde{x}|\theta,x)=p(\tilde{x}|\theta)$. **Indicator:** $\mathbb{E}[I(A)]=P(A)$.

Tower: $\mathbb{E}X=\mathbb{E}[\mathbb{E}(X|Z)]$. **Total variance:** $\text{Var}X=\mathbb{E}[\text{Var}(X|Z)]+\text{Var}[\mathbb{E}(X|Z)]$. **Linear:** $\text{Var}(aX+bY)=a^2\text{Var}X+b^2\text{Var}Y+2ab\text{Cov}(X,Y)$; $\text{Var}(t^\top\hat{\theta})=t^\top Vt$; $\text{Var}(AY)=A\text{Var}(Y)A^\top$.

Square trick (CAVI): $\mathbb{E}[(c-\mu)^2]=(c-\mathbb{E}\mu)^2+\text{Var}\mu$, so $\mathbb{E}[\mu^2]=(\mathbb{E}\mu)^2+\text{Var}\mu$.

Jensen: f convex $\Rightarrow \mathbb{E}f(X)\ge f(\mathbb{E}X)$. Log is concave: $\log\mathbb{E}Y\ge\mathbb{E}\log Y$, equal iff Y is constant.

Invariance: MLE of $g(\theta)$ is $g(\hat{\theta})$. **Delta:** $g(\hat{\theta})\approx \text{N}(g(\theta),g'(\theta)^2I(\theta)^{-1})$. **Taylor:** $g(Y)\approx g(\mu)+(Y-\mu)g'(\mu)$.

Integrals: $\int_0^\infty x^{a-1}e^{-bx}dx=\Gamma(a)/b^a$; $\int_0^1 x^{a-1}(1-x)^{b-1}dx=B(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$; $\Gamma(a+1)=a\Gamma(a)$; $\Gamma(n)=(n-1)!$; $\Gamma(\frac{1}{2})=\sqrt{\pi}$. **Normal integral:** $\int e^{-\frac{A}{2}\mu^2+B\mu}d\mu=\sqrt{2\pi/A}\,e^{B^2/2A}$.

Inverse: $\begin{pmatrix}a&b\\c&d\end{pmatrix}^{-1}=\frac{1}{ad-bc}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$. **Odds:** $o=\frac{p}{1-p}$, $p=\frac{o}{1+o}$.

Complete the square: $-\frac{A}{2}\mu^2+B\mu$ is the kernel of $\text{N}(B/A,\ 1/A)$. Course log = ln: press **ln** on the FX-82, not log. Round final answers to 2dp. FX-82: type $(-0.5)^2$ with brackets; $\sum\log a_i$ is one **ln** of the product.

MIXTURES & EM

Mixture model: (1) $Z_i\sim\text{Categorical}(\pi_1,\dots,\pi_K)$, Z_i latent label, π_k mixture proportions. (2) $X_i|Z_i=k\sim P(X_i|\Theta_k)$ component. $P(x_i)=\sum_k\pi_k f(x_i;\Theta_k)$, $L(\theta)=\prod_i\sum_k\pi_k f(x_i;\Theta_k)$. Free parameters: $K-1$ proportions + component parameters. A mixture is not a linear combination: $\pi_aX_a+\pi_bX_b$ of Normals is Normal; a mixture may be bimodal, or unimodal and skewed.

Mean: $\bar{\mu}=\sum\pi_k\mu_k$. **Var:** $\sum\pi_k\sigma_k^2+\sum\pi_k(\mu_k-\bar{\mu})^2$. Poisson mixtures are overdispersed.

Clustering: $P(Z_i=k|X_i)=\frac{\pi_k f(x_i;\Theta_k)}{\sum_j\pi_j f(x_i;\Theta_j)}$; assign to the largest. Common factors cancel ($x!$, $\sqrt{2\pi}$) but not π_k or unequal $1/\sigma_k$. Poisson, $K=2$: $P_{i1}=1/[1+\frac{\pi_2}{\pi_1}(\frac{\lambda_2}{\lambda_1})^{x_i}e^{-(\lambda_2-\lambda_1)}]$. Normal, equal σ^2, $\mu_1<\mu_2$: cluster 1 if $x<\frac{\mu_1+\mu_2}{2}+\frac{\sigma^2\log(\pi_1/\pi_2)}{\mu_2-\mu_1}$.

EM algorithm: $l(\theta)=\sum_i\log\sum_k\pi_k f$: the sum inside the log blocks a closed form. If Z were known it would be easy. **Incomplete log-lik:** $\log P(X|\theta)=\log\sum_Z P(X,Z|\theta)$. **Complete:** $\log P(X,Z|\theta)$. **Q:** $Q(\theta,\theta^0)=\mathbb{E}_{Z|X,\theta^0}[\log P(X,Z|\theta)]$.

Steps: (1) Set θ^0; evaluate incomplete log-lik. (2) **E:** compute $P(Z|X,\theta^0)$. (3) **M:** $\hat{\theta}=\arg\max_\theta Q(\theta,\theta^0)$. (4) Evaluate incomplete log-lik; stop if change $<\epsilon$, else $\theta^0\leftarrow\hat{\theta}$, go to 2.

Deriving Q: write $P_{ik}=P(Z_i=k|X_i,\theta^0)$.

$$P(X,Z|\theta)=\prod_i\prod_k[\pi_k f(x_i;\Theta_k)]^{I(Z_i=k)}$$

$$\log P(X,Z|\theta)=\sum_i\sum_k I(Z_i=k)[\log\pi_k+\log f(x_i;\Theta_k)]$$

$\mathbb{E}[I(Z_i=k)]=P_{ik}$ and only I is random, so

$$Q=\sum_i\sum_k P_{ik}[\log\pi_k+\log f(x_i;\Theta_k)]$$

$K=2$: $\pi_1=\pi$, $\pi_2=1-\pi$. Normal: $\log f=-\frac{1}{2}\log(2\pi\sigma_k^2)-\frac{(x_i-\mu_k)^2}{2\sigma_k^2}$ (that π is 3.14). Poisson: $x_i\log\lambda_k-\lambda_k-\log x_i!$. Normal answer in full:

$$Q=\sum_i\sum_{k=1}^2 P_{ik}[\log\pi_k-\frac{1}{2}\log2\pi\sigma_k^2-\frac{(x_i-\mu_k)^2}{2\sigma_k^2}]$$

M-step, $N_k=\sum_i P_{ik}$ (effective count):

- π: $\frac{\partial Q}{\partial\pi}=\frac{N_1}{\pi}-\frac{N_2}{1-\pi}=0$, so $\hat{\pi}_k=N_k/n$. $K\ge3$: Lagrange $Q+\nu(1-\sum\pi_k)$ gives $\pi_k=N_k/\nu$; sum over k: $\nu=n$.
- **Normal:** $\hat{\mu}_k=\sum_i P_{ik}x_i/N_k$, $\hat{\sigma}_k^2=\sum_i P_{ik}(x_i-\hat{\mu}_k)^2/N_k$ (from $\frac{\partial Q}{\partial\sigma^2}=\sum P_{ik}[-\frac{1}{2\sigma^2}+\frac{(x_i-\mu)^2}{2\sigma^4}]=0$). Mean 0: $\sum P_{ik}x_i^2/N_k$. Shared σ^2: $\sum_i\sum_k P_{ik}(x_i-\hat{\mu}_k)^2/n$.
- **Poisson:** $\hat{\lambda}_k=\sum P_{ik}x_i/N_k$; $Q''=-\sum P_{ik}x_i/\lambda_k^2<0$. **Bin**$(m,p_k)$: $\hat{p}_k=\sum P_{ik}x_i/(mN_k)$. **Exp(rate λ_k):** $\hat{\lambda}_k=N_k/\sum P_{ik}x_i$.

Twists rule: collect every term with the parameter across all components and blocks, differentiate once. Exp(λ), Exp(2λ): $\hat{\lambda}=n/\sum(P_{i1}+2P_{i2})x_i$. Exposure t_i: $\hat{\lambda}_k=\sum P_{ik}x_i/\sum P_{ik}t_i$. $\text{N}(\mu,\sigma^2)$, $\text{N}(\mu,9\sigma^2)$: $w_i=P_{i1}+P_{i2}/9$, $\hat{\mu}=\sum w_ix_i/\sum w_i$, $\hat{\sigma}^2=\sum w_i(x_i-\hat{\mu})^2/n$.

n' **points known from component** j (not drawn from the mixture): no Z, so they add $\sum\log f_j(x_i)$ to Q with weight 1 and no π. $\hat{\pi}_k=N_k/n$, unlabelled n only. $\hat{\mu}_j=(\sum P_{ik}x_i+\sum_{lab}x_i)/(N_j+n')$; same for σ_j^2 with $(x_i-\hat{\mu}_j)^2$. Incomplete log-lik: $\sum_{unlab}\log\sum_k\pi_kf_k+\sum_{lab}\log f_j$.

Observed labels from the mixture: count in both: $\hat{\pi}_k=(N_k+m_k)/(n+m)$. **Known parameters:** do not differentiate. **Tied:** $\text{N}(\mu,1)$,$\text{N}(-\mu,1)$: differentiate once, $\hat{\mu}=\sum(P_{i1}-P_{i2})x_i/n$. **Zero-inflated:** $P_{i1}=0$ if $x_i>0$; $\hat{\lambda}=\sum x_i/(n-N_1)$. $K=3$: $\pi_3=1-\pi_1-\pi_2$, three terms in the denominator.

Behaviour: incomplete log-lik **never decreases** (a drop means a bug). Local maxima: sensitive to initial values, so run many starts, keep the **highest incomplete log-lik**. **Label switching:** same log-lik, permuted labels, same fit. Equal starts ($\lambda_1^0=\lambda_2^0$) get stuck at $\hat{\lambda}_k=\bar{x}$, π unchanged. Overlap: P_{ik} less decisive, EM slower and more start-sensitive. Do not compare log-lik across different data.

By hand: table of x_i, P_{i1}, P_{i2}; check $N_1+N_2=n$. M-step is the complete-data MLE with hard counts replaced by weights P_{ik}. To choose between runs, plug each $\hat{\theta}$ into $\sum_i\log(\pi f_1(x_i)+(1-\pi)f_2(x_i))$. Keep $\log x_i!$ when reporting Q or log-lik values.

Proof (Jensen): $q(Z)=P(Z|X,\theta^0)$:

$$l(\theta)=\log\sum_Z q\frac{P(X,Z|\theta)}{q}\ge\sum_Z q\log\frac{P(X,Z|\theta)}{q}$$

$=Q(\theta,\theta^0)+C$, $C=-\sum q\log q$. Equality at θ^0, as $P(X,Z|\theta^0)/q=P(X|\theta^0)$ is constant. M-step: $Q(\hat{\theta},\theta^0)\ge Q(\theta^0,\theta^0)$. So $l(\hat{\theta})\ge Q(\hat{\theta},\theta^0)+C\ge Q(\theta^0,\theta^0)+C=l(\theta^0)$.

EM shortcuts: write "the terms of Q involving λ are…" before each derivative.

Any two classes, no underflow: $P_{i1}=1/(1+e^d)$, $d=\log\frac{\pi_2\sigma_1}{\pi_1\sigma_2}+\frac{(x-\mu_1)^2}{2\sigma_1^2}-\frac{(x-\mu_2)^2}{2\sigma_2^2}$.

- Pois(λ), Pois(3λ): $P_{i1}=1/[1+\frac{1-\pi}{\pi}3^{x_i}e^{-2\lambda}]$; $\hat{\lambda}=\sum x_i/\sum(P_{i1}+3P_{i2})$, known block adds 3 each below.
- Pois($t\lambda_1$), Pois($t\lambda_2$): $P_{i1}=1/[1+\frac{1-\pi}{\pi}(\frac{\lambda_2}{\lambda_1})^{x_i}e^{-t_i(\lambda_2-\lambda_1)}]$.
- $\text{N}(\mu,\sigma^2)$, $\text{N}(-\mu,\sigma^2)$: $P_{i1}=\pi/[\pi+(1-\pi)e^{-2\mu x_i/\sigma^2}]$; $\hat{\mu}=\frac{1}{n}\sum(2P_{i1}-1)x_i$; $\hat{\sigma}^2=\frac{1}{n}\sum x_i^2-\hat{\mu}^2$.
- Exp(λ), Exp(2λ): $P_{i1}=\pi/[\pi+2(1-\pi)e^{-\lambda x_i}]$.
- Zero-inflated Poisson: $x_i=0$: $P_{i1}=\pi/[\pi+(1-\pi)e^{-\lambda}]$; $x_i>0$: 0. Binomial: $\pi/[\pi+(1-\pi)(1-p)^{m_i}]$; $\hat{p}=\sum x_i/\sum P_{i2}m_i$.
- $\text{N}(\mu,1)$, $\text{N}(2\mu,4)$: $\hat{\mu}=\frac{1}{n}\sum(P_{i1}+\frac{1}{2}P_{i2})x_i$.

Mixture moments: $K=2$ between term $=\pi_1\pi_2(\mu_1-\mu_2)^2$. Poisson: $\text{Var}X=\bar{\lambda}+\pi_1\pi_2(\lambda_1-\lambda_2)^2$, so intercept-only $\hat{\phi}\approx\text{Var}X/\mathbb{E}X$. Exposure t: $t\bar{\lambda}+t^2\pi_1\pi_2(\lambda_1-\lambda_2)^2$. Zero-inflated: $\mathbb{E}X=(1-\pi)\lambda$, $\text{Var}X=\mathbb{E}X(1+\pi\lambda)$.

Cutoffs: Poisson, $\lambda_1<\lambda_2$: cluster 1 if $x<\frac{\log(\pi_1/\pi_2)+\lambda_2-\lambda_1}{\log(\lambda_2/\lambda_1)}$. Same mean μ, $\sigma_1<\sigma_2$: cluster 2 if $|x-\mu|>c$, $c^2=\log\frac{\pi_1\sigma_2}{\pi_2\sigma_1}/(\frac{1}{2\sigma_1^2}-\frac{1}{2\sigma_2^2})$ (log ≤0: all cluster 2). Misclassified: $\pi_1P(x>c|1)+\pi_2P(x\le c|2)$.

By hand: $\sum P_{i2}x_i=\sum x_i-\sum P_{i1}x_i$. Pooled $\hat{\sigma}^2=(N_1\hat{\sigma}_1^2+N_2\hat{\sigma}_2^2)/n$. Variance update uses the new mean. $K=3$: $N_k/\pi_k=c$ for all k, sum gives $c=n$.

Is it a maximum? $Q''_\mu=-N_k/\sigma_k^2$; $Q''_\pi=-N_1/\pi^2-N_2/(1-\pi)^2$; $Q''_{\sigma^2}=-N_k/2\hat{\sigma}_k^4$.

Choose K: l always rises with K: use AIC, $p=K-1$ + component parameters (known values not counted).

Maid Mast Codes PT2: EM

`Z=sample(1:K,n,replace=TRUE,prob=pi); X=rnorm(n,mu[Z],sd[Z])`: simulate.

- `L[,k]=dnorm(X,mu[k],s)*w[k]`: $n\times K$ matrix $P(X_i,Z_i{=}k)$.
- `P_ik=prob.x.z/rowSums(prob.x.z)`: E-step.
- `w.new=colSums(P_ik)/sum(P_ik)`: $\hat{\pi}_k$; `sum(P_ik)` $=n$.
- `mu.new=colSums(P_ik*X)/colSums(P_ik)`: $\hat{\mu}_k$.
- `ill=sum(log(rowSums(prob.x.z)))`: incomplete log-lik.
- `delta.ll`, `epsilon`: gain, tolerance. Negative `delta.ll` also stops the loop, so a bug can look converged. Bugs: `/colSums` in E-step; `sum(log(prob.x.z))`.
- `cluster=apply(P_ik,1,which.max)`: largest P_{ik}. `table(true=Z,cluster)`: diagonal correct, needs simulated labels; permute columns if labels switched.
- `length(log_liks)` = iterations +1; = `max.iter+1`: stopped at the cap, not converged, rerun longer. All means equal: every row of `P_ik` is π^0, stuck at $\bar{x}$ after 2 iterations.
- **More bugs:** `pi.new=colSums(P_ik)`: π sums to n, l jumps by $n\log n$ yet still rises. `mu.new=colSums(P_ik*X)/sum(P_ik)`: l can fall, loop stops, looks converged.
- **Say:** l rises every iteration and levels off, so it has converged.

BAYES

Bayes: Frequentist: θ fixed, probability = long-run frequency. Bayesian: θ random, probability = degree of belief.

$$p(\theta|x)=\frac{f(x|\theta)p(\theta)}{f(x)}\propto f(x|\theta)p(\theta)$$

Likelihood × prior. Marginal $f(x)=\int f(x|\theta)p(\theta)d\theta$. **Point estimate:** posterior mean (minimises squared error loss). **Credible interval:** central 95% of the posterior, a direct probability statement about θ. Normal posterior: mean ±1.96 sd; samples: `quantile(th,c(.025,.975))`.

Kernels: **Gamma**(a,b): $\tau^{a-1}e^{-b\tau}$. **InvGamma**(a,b): $x^{-a-1}e^{-b/x}$. **Normal:** $e^{-\frac{A}{2}\mu^2+B\mu}$ is $\text{N}(B/A,1/A)$. **Beta**(a,b): $\theta^{a-1}(1-\theta)^{b-1}$.

In λ: Poisson likelihood is $\lambda^{\sum y_i}e^{-n\lambda}$, Exp is $\lambda^ne^{-\lambda\sum y_i}$: Gamma kernels. Exp(rate b) = Gamma$(1,b)$.

Recipe: joint → keep factors with the parameter → add exponents → name the kernel. Summaries: $\sum(x_i-c)^2=\sum(x_i-\bar{x})^2+n(\bar{x}-c)^2$. $\tau^{n/2}$ is constant for μ but not for τ: keep it. Each prior appears once.

Likelihood (n)	Prior	Posterior
Bin(m,θ)	Beta	Beta$(a+\sum y_i,\ b+nm-\sum y_i)$
Geom ($x\ge1$)	Beta	Beta$(a+n,\ b+\sum y_i-n)$
Poisson(λ)	Gamma	Gamma$(a+\sum y_i,\ b+n)$
Exp(rate λ)	Gamma	Gamma$(a+n,\ b+\sum y_i)$
$\text{N}(\mu_0,1/\tau)$	Gamma	Gamma$(a+\frac{n}{2},\ b+\frac{1}{2}S)$
$\text{N}(\mu_0,\sigma^2)$	InvG	InvGamma$(a+\frac{n}{2},\ b+\frac{1}{2}S)$

$S=\sum(x_i-\mu_0)^2$. $\mathbb{E}[\sigma^2|x]=b_n/(a_n-1)$, not $1/\mathbb{E}[\tau|x]=b_n/a_n$. Poisson($t_i\lambda$): rate $b+\sum t_i$.

Normal mean, σ^2 known: prior $\text{N}(\mu,\sigma^2/n_0)$ (n_0 = implicit sample size):

$$\theta|x\sim\text{N}\Big(\frac{n_0\mu+n\bar{x}}{n_0+n},\ \frac{\sigma^2}{n_0+n}\Big)$$

Prior $\text{N}(m,s^2)$: precisions add: variance $(\frac{1}{s^2}+\frac{n}{\sigma^2})^{-1}$, mean = variance $\times(\frac{m}{s^2}+\frac{n\bar{x}}{\sigma^2})$. $\bar{X}\sim\text{N}(m,\frac{\sigma^2}{n}+s^2)$ marginally.

Posterior mean is a precision-weighted average of prior mean and MLE. Variance is below both prior and MLE variance. $n\to\infty$: $\to\text{N}(\bar{x},\sigma^2/n)$, prior forgotten. $n_0\to0$: flat prior. Weight on the prior mean $w=\kappa_0/(\kappa_0+n\tau)=n_0/(n_0+n)$; $\kappa_0\approx n\tau$ gives the prior half the say.

Posterior predictive: $p(\tilde{x}|x)=\int p(\tilde{x}|\theta)p(\theta|x)d\theta$. Normal: N(post. mean, post. var + σ^2). Simulate: $\theta^{(l)}\sim p(\theta|x)$, then $\tilde{x}^{(l)}\sim p(\tilde{x}|\theta^{(l)})$; average. Known mean μ_0, $\tau|x\sim\text{Gamma}(a^*,b^*)$: 3-parameter t with $\nu=2a^*$, location μ_0, scale² $s^2=b^*/a^*$, pdf $\propto(1+\frac{(x-\mu_0)^2}{\nu s^2})^{-(\nu+1)/2}$. Poisson-Gamma: $\text{Var}\tilde{y}=\mathbb{E}[\lambda|y]+\text{Var}(\lambda|y)$; $P(\tilde{y}=0)=(\frac{b_n}{b_n+1})^{a_n}$.

Pseudo-counts: Beta(a,b) acts like $a+b$ prior trials; Beta(1,1) = U(0,1). Poisson-Gamma mean: $\frac{a+\sum y_i}{b+n}=w\frac{a}{b}+(1-w)\bar{y}$, $w=\frac{b}{b+n}$. $\text{Var}(\lambda|y)=a_n/b_n^2$. Exp-Gamma mean: weight $\frac{b}{b+\sum x_i}$ on a/b, rest on $n/\sum x_i$. Gamma interval: mean ±1.96 sd is a shortcut; exact uses quantiles.

Marginals: Beta-binomial $P(X{=}x)=\binom{m}{x}B(a+x,b+m-x)/B(a,b)$; Poisson-Gamma gives NegBin. Poisson-Gamma $p(y)=\frac{\Gamma(a+y)}{\Gamma(a)y!}\frac{b^a}{(b+1)^{a+y}}$, $\text{Var}=\mu+\mu^2/a$. X_1,X_2 iid only given λ: $\text{Cov}=\text{Var}\lambda>0$.

Priors: vague $\text{N}(0,10^4)$, Gamma(.001,.001); **informative**: concentrated. **Improper:** does not integrate to 1, $p(\mu)\propto1$, $p(\tau)\propto1/\tau$ (Gibbs: $a=b=0$); fine if the posterior is proper ($p(\lambda)\propto1/\lambda$, Poisson: Gamma$(\sum y_i,n)$, needs $\sum y_i>0$). $p(\mu,\sigma^2)\propto1/\sigma^2$ goes straight into π in MH. Flat μ, τ known: $\text{N}(\bar{x},\frac{1}{n\tau})$. $p(\tau)\propto1/\tau$, mean known: Gamma$(\frac{n}{2},\frac{S}{2})$. Flat μ and $1/\tau$: $\tau|x\sim\text{Gamma}(\frac{n-1}{2},\frac{1}{2}\sum(x_i-\bar{x})^2)$, needs $n\ge2$. $x_i\sim\text{N}(0,\sigma^2)$, $p(\sigma^2)\propto1/\sigma^2$: InvGamma$(\frac{n}{2},\frac{1}{2}\sum x_i^2)$, mean $\sum x_i^2/(n-2)$ if $n\ge3$.

BUGS: `dnorm(m,p)`: second argument is a precision: `mu~dnorm(0,.001)` is $\text{N}(0,1000)$, vague. Vague prior: expect mean $\approx\bar{y}$, sd $\approx1/\sqrt{n\tau}$.

Spike prior: γ discrete: $p(\gamma|x)\propto p(x|\gamma)p(\gamma)$, with $p(x|\gamma)=\int p(x|\mu)p(\mu|\gamma)d\mu$; sample γ, then $\mu|\gamma,x$.

Conjugate exp family: $f=\alpha(x)e^{x\theta+\beta(\theta)}$, prior $\propto e^{\gamma\theta+\delta\beta(\theta)}$: posterior $(\gamma+x,\delta+1)$.

Bayes estimator: minimise posterior expected loss; squared error gives posterior mean. **Bayesian GLM:** vague priors $\beta_j\sim\text{N}(0,10^4)$; overdispersion: add $\varepsilon_i\sim\text{N}(0,1/\tau)$ to η_i, or $\lambda_i\sim$ Gamma per unit. **Joint prior** $\mu|\tau$, τ: exact posterior (VI box), so sample τ then μ, no MCMC.

SIMULATION

Simulation: **Monte Carlo:** $\frac{1}{N}\sum g(X_j)\approx\mathbb{E}g(X)$; proportions estimate probabilities. `set.seed` makes it repeatable. Need $N\ge\hat{p}(1-\hat{p})/\text{se}^2$ (at most $.25/\text{se}^2$). sd of draws is the posterior spread, fixed; MC s.e. = sd$/\sqrt{N}$ shrinks.

Discrete: $U\sim$ U(0,1); `x<-0; while(F(x)<U) x<-x+1`. Ends with $F(x-1)<U\le F(x)$, so $P(X=x)=F(x)-F(x-1)=p(x)$. **Read a pmf from a loop:** `while(U>G(Y)) Y<-Y+1` means $F=G$, then difference it. `sample(x,n,replace=TRUE,prob=p)`. Expected comparisons: $\sum$ (tests needed) × prob, so test likely values first.

Inversion: $Y=F^{-1}(U)$: $P(Y\le y)=P(U\le F(y))=F(y)$. Needs F^{-1}. Exp(λ): $-\log(1-U)/\lambda$; U(a,b): $a+(b-a)U$; logistic: $\log\frac{U}{1-U}$; Weibull: $(-\log U)^{1/k}/\lambda$. Pdf $2x$ on (0,1): $\sqrt{U}$. e^{-z} on (0,1): $-\log(1-U(1-e^{-1}))$.

Rejection, uniform: $f\le k$ on $[a,b]$. (1) $X\sim$ U(a,b), $Y\sim$ U$(0,k)$. (2) If $Y<f(X)$ return X, else go to 1.

General: envelope density h, $k\ge\sup_x f(x)/h(x)$. (1) $X\sim h$. (2) $Y\sim$ U$(0,kh(X))$. (3) If $Y<f(X)$ return X, else go to 1. Works with unnormalised f^* (posterior): $k\ge\sup f^*/h$. Prior as h: $k=\max_\theta p(x|\theta)$, keep if $U\le p(x|\theta)/k$. k too small biases the output (where $f>kh$ is under-sampled); too big only slows. h tails at least as heavy as f.

Why: (X,Y) is uniform under kh (density $1/k$); points kept are uniform under f, so $X\sim f$.

Derive output pdf: $P(X\le x|\text{accept})=P(X\le x,\text{accept})/P(\text{accept})=\int_{-\infty}^x h(t)\frac{f(t)}{kh(t)}dt/\frac{1}{k}=F(x)$.

Efficiency: expected candidates per sample. $P(\text{accept})=1/k$ (with f^*: $\int f^*/k$). $N\sim$ Geom$(1/k)$, $\mathbb{E}N=k$ (f^*: $k/\int f^*$). Least rejection: smallest valid k, h close to f. Uniform envelope: rectangle height $\sup f$ on $[a,b]$, so $k=(b-a)\sup f$. Then accept if $U\le f^*(X)/\max f^*$.

Gamma(shape m, rate λ), $m>1$, $h=\mu e^{-\mu x}$: f/h peaks at $x=\frac{m-1}{\lambda-\mu}$. Best $\mu=\lambda/m$, $k^*=m^me^{-(m-1)}/\Gamma(m)$. $k^*=\infty$ if $m<1$ or $\lambda\le\mu$.

Maid Mast Codes PT3: Simulation

- `x<-runif(1,a,b); y<-runif(1,0,K); if(y<fx(x)) return(x)`: rejection.
- `X<- -log(runif(1))*m/lambda`: Exp(rate λ/m) by inversion.
- `tha<-rbeta(N,2,3); X<-rbinom(N,10,tha); table(X)/N`: marginal pmf by simulating parameter then data.
- `while(runif(1)>p) Y<-Y+1` from $Y=0$: failures before first success, $P(Y=y)=p(1-p)^y$.
- `xt<-rbinom(N,10,theta)`: each draw uses its own θ, so `mean(xt>=8)` is predictive. Label then x is a mixture draw.

MCMC

MCMC: build a Markov chain whose stationary distribution is the posterior. Long-run draws are dependent but approximately from the posterior.

Metropolis-Hastings: target $\pi\propto p(\theta)p(x|\theta)$ (constant not needed). **Proposal** $q(\theta,\theta')$: density of new θ' given current θ; must be simulable.
(1) Start $\theta^{(0)}$, $t=0$. (2) Propose $\theta'\sim q(\theta^{(t)},\cdot)$. (3)

$$A=\min\Big(1,\ \frac{\pi(\theta')q(\theta',\theta)}{\pi(\theta)q(\theta,\theta')}\Big)$$

(4) With probability A set $\theta^{(t+1)}=\theta'$, else $\theta^{(t+1)}=\theta$ ($U\le A$). (5) $t\leftarrow t+1$, go to 2.

Random walk: $\theta'=\theta+$ symmetric noise (U$[-\epsilon,\epsilon]$, $\text{N}(0,\epsilon^2)$): q cancels, $A=\min(1,\pi(\theta')/\pi(\theta))$. Uphill always accepted, downhill sometimes, so time spent $\propto\pi$. **Independence:** $q(\theta,\theta')=q(\theta')$, so $A=\min\{1,w(\theta')/w(\theta)\}$, $w=\pi/q$. **Walk on** $\log\tau$**:** extra factor τ'/τ.

Numeric A recipe: (i) π = likelihood × prior at new and current. (ii) q ratio: top is density of proposing current from new, bottom new from current. Multi-part proposals multiply: $\tau_n\sim g(\cdot|\tau_c)$, $\mu_n\sim h(\cdot|\mu_c,\tau_n)$ gives $q(c,n)=g(\tau_n|\tau_c)h(\mu_n|\mu_c,\tau_n)$, $q(n,c)=g(\tau_c|\tau_n)h(\mu_c|\mu_n,\tau_c)$. (iii) Cancel common constants only. (iv) $\min(1,\cdot)$. Out of support: $\pi=0$, $A=0$. Log form (symmetric q): `A<-min(1,exp(lp(new)-lp(cur)))`; else add log q ratio.

Ledger: one line each: likelihood (keep $\tau^{n/2}$), each prior (keep $\sqrt{\tau}$), each proposal, multiply, $\min(1,\cdot)$. A correct unfinished ledger still scores.

Proposal ratios $q(c|n)/q(n|c)$: Gamma(k, rate k/τ_c): $(\tau_c/\tau_n)^{2k-1}e^{-k\tau_c/\tau_n+k\tau_n/\tau_c}$ ($k=1$: Exp, mean τ_c). Exp(rate τ_c): τ_n/τ_c. $\mu_n\sim\text{N}(\mu_c,1/\tau_n)$: $\sqrt{\tau_c/\tau_n}\,e^{-\frac{1}{2}(\tau_c-\tau_n)(\mu_n-\mu_c)^2}$. Walk on $\log\tau$: τ_n/τ_c. Same variance both ways: 1.

Symmetric? Swap current and new: same formula for every pair. $\text{N}(\theta_c,s^2)$ yes; $\text{N}(0,1)$, Exp(rate $1/\theta_c$) no. Leave q out and it runs silently on the wrong target. `dunif` prior out of range: log is $-$Inf, rejected. Normal step on τ can go negative: $\pi=0$, wasted. Independence: q tails as heavy as π, else sticky.

Tuning: want large moves and high acceptance; they compete, so aim for intermediate A. Tiny steps: high acceptance, slow crawl. Huge steps: low acceptance, sticky. Both: high autocorrelation, low ESS. Rule of thumb (not slides): 20–50% accepted.

Read π off A: symmetric q and $A=\min(1,r(\theta_n)/r(\theta_o))$ give $\pi\propto r$.

Why it works (not examinable): detailed balance $\pi(\theta)p(\theta\to\theta')=\pi(\theta')p(\theta'\to\theta)$ makes π stationary; aperiodic, irreducible: limiting; ergodic: averages converge.

Derivation, $x\sim\text{N}(\mu,1)$, $\mu\sim\text{N}(0,1)$, random walk: $\log\frac{\pi(\mu')}{\pi(\mu)}=-\frac{1}{2}[(x-\mu')^2+\mu'^2]+\frac{1}{2}[(x-\mu)^2+\mu^2]$; $A=\min(1,e^{\text{this}})$.

Gibbs sampler: needs only full conditionals. $\theta=(\theta_1,\dots,\theta_k)$. Start $\theta^{(0)}$. Each sweep: $\theta_j^{(t+1)}\sim p(\theta_j|x,\text{latest values of others})$, $j=1,\dots,k$. Discard burn-in m; keep the rest as a dependent sample; marginals: keep that coordinate.

Use output: $\mathbb{E}[g(\theta)|x]\approx$ average of $g(\theta^{(t)})$ after burn-in; $P(\theta>c|x)\approx$ proportion above c; credible interval from sample quantiles. Apply g to each draw, then average: $\mathbb{E}[\mu\theta]$ = mean of products; $\mathbb{E}\sigma\ne1/\sqrt{\mathbb{E}\tau}$; mean of ratios ≠ ratio of means. Difference: $\delta^{(t)}=\mu_1^{(t)}-\mu_2^{(t)}$, then `mean(d<0)`.

Special case of MH: proposal = full conditional, so $A=1$. **Limit:** strong correlation between coordinates gives tiny moves and slow exploration. Bivariate, correlation ρ: lag-1 autocorrelation of one coordinate is ρ^2.

Not a named distribution? Finite support: sample with weights (R normalises `prob`). Otherwise an MH step for that coordinate.

Check before dropping a prior: does it contain the parameter you condition on? Scheme answer: start values ($\tau^{(0)}>0$), each draw with the newest values, drop burn-in, average.

Standard conditionals ($x_i\sim\text{N}(\mu,1/\tau)$, $S=\sum(x_i-\mu)^2$):

- $\mu\sim\text{N}(m,\frac{1}{\kappa})$: $\mu|\tau,x\sim\text{N}(\frac{\kappa m+\tau n\bar{x}}{\kappa+n\tau},\frac{1}{\kappa+n\tau})$.
- $\mu|\tau\sim\text{N}(m,\frac{1}{\kappa\tau})$: $\mu|\tau,x\sim\text{N}(\frac{\kappa m+n\bar{x}}{\kappa+n},\frac{1}{(\kappa+n)\tau})$.
- $\tau\sim\text{Gamma}(a,b)$: $\tau|\mu,x\sim\text{Gamma}(a+\frac{n}{2},b+\frac{S}{2})$; with the $\mu|\tau$ prior: Gamma$(a+\frac{n+1}{2},b+\frac{S}{2}+\frac{\kappa}{2}(\mu-m)^2)$.
- $\sigma^2\sim\text{InvGamma}(a,b)$: InvGamma$(a+\frac{n}{2},b+\frac{S}{2})$; prior $\mu|\sigma^2\sim\text{N}(m,\sigma^2/\kappa)$: InvGamma$(a+\frac{n+1}{2},b+\frac{S}{2}+\frac{\kappa}{2}(\mu-m)^2)$.
- **Two samples** x (n_1), y (n_2), common τ: Gamma$(a+\frac{n_1+n_2}{2},b+\frac{S_x+S_y}{2})$; μ_1 uses x only.
- **One μ, two sources:** $x_i\sim\text{N}(\mu,\frac{1}{\tau})$ (n), $y_j\sim\text{N}(c\mu,\frac{1}{d\tau})$ (m), $\mu\sim\text{N}(\mu_0,\frac{1}{\kappa})$: $A=\kappa+(n+c^2dm)\tau$, $B=\kappa\mu_0+\tau(\sum x_i+cd\sum y_j)$; $\tau|\cdot\sim\text{Gamma}(a+\frac{n+m}{2},b+\frac{1}{2}[\sum(x_i-\mu)^2+d\sum(y_j-c\mu)^2])$.
- **Hierarchical:** $y_{ij}\sim\text{N}(\theta_j,\frac{1}{\tau})$, $\theta_j\sim\text{N}(\mu,\frac{1}{\tau_\theta})$, $\mu\sim\text{N}(m,\frac{1}{\kappa})$: $\theta_j|\cdot$ precision $n_j\tau+\tau_\theta$, mean $(n_j\tau\bar{y}_j+\tau_\theta\mu)$/prec (weight $n_j\tau$/prec on $\bar{y}_j$); $\mu|\cdot$ precision $\kappa+J\tau_\theta$, mean $(\kappa m+\tau_\theta\sum\theta_j)$/prec (no y); $\tau_\theta|\cdot\sim\text{Gamma}(a+\frac{J}{2},b+\frac{1}{2}\sum(\theta_j-\mu)^2)$.
- **Bivariate prior**, $\Lambda=\Sigma^{-1}$, zero mean, $x_i\sim\text{N}(\mu_1,\sigma^2)$: $-\frac{1}{2}(\Lambda_{11}\mu_1^2+2\Lambda_{12}\mu_1\mu_2+\Lambda_{22}\mu_2^2)$ in the exponent gives $A=\Lambda_{11}+n/\sigma^2$, $B=n\bar{x}/\sigma^2-\Lambda_{12}\mu_2$: $\mu_1|\mu_2,x\sim\text{N}(B/A,1/A)$.
- **Change point:** $Y_i\sim$ Pois(λ_1), $i\le m$; Pois(λ_2), $i>m$; Gamma priors; m uniform on 1..n: $\lambda_1|\cdot\sim\text{Gamma}(\alpha_1+\sum_1^m y_i,\beta_1+m)$; $\lambda_2|\cdot\sim\text{Gamma}(\alpha_2+\sum_{m+1}^n y_i,\beta_2+n-m)$; $p(m|\cdot)\propto(\lambda_1/\lambda_2)^{\sum_1^m y_i}e^{(\lambda_2-\lambda_1)m}$.
- **Exp change point:** $\lambda_1|\cdot\sim\text{Gamma}(a_1+M,b_1+S_M)$, $\lambda_2|\cdot\sim\text{Gamma}(a_2+n-M,b_2+S_n-S_M)$, $S_m=\sum_1^m y_i$; $p(m|\cdot)\propto(\lambda_1/\lambda_2)^me^{-(\lambda_1-\lambda_2)S_m}$.
- **Hidden count:** $X|p,N\sim\text{bin}(N,p)$, $p\sim\text{Beta}(a,b)$, $N\sim$ Pois(λ): $p|\cdot\sim\text{Beta}(x+a,N-x+b)$; $N-x|\cdot\sim\text{Pois}(\lambda(1-p))$.
- **Discrete** $\mu\in\{0,1\}$: $P(\mu=1|\theta)=f(\theta,1)/[f(\theta,0)+f(\theta,1)]$.
- **Two groups, flat priors,** $p(\tau)\propto1/\tau$: $\mu_1|\cdot\sim\text{N}(\bar{x},\frac{1}{n\tau})$, $\mu_2|\cdot\sim\text{N}(\bar{y},\frac{1}{m\tau})$, $\tau|\cdot\sim\text{Gamma}(\frac{n+m}{2},\frac{1}{2}[\sum(x_i-\mu_1)^2+\sum(y_j-\mu_2)^2])$.
- **Poisson two sources:** $y_i\sim$ Pois(λ) (n), $z_j\sim$ Pois($c\lambda$) (m): $\lambda|\cdot\sim\text{Gamma}(a+\sum y_i+\sum z_j,b+n+cm)$.
- **Poisson hierarchy:** $\lambda_j|\beta\sim\text{Gamma}(\alpha,\beta)$, $\beta\sim\text{Gamma}(c,d)$: $\lambda_j|\cdot\sim\text{Gamma}(\alpha+\sum_i y_{ij},\beta+n_j)$; $\beta|\cdot\sim\text{Gamma}(c+J\alpha,d+\sum\lambda_j)$.

Derivation, Gibbs μ: $p(\mu|\tau,x)\propto\exp\{-\frac{\tau}{2}\sum(x_i-\mu)^2-\frac{\kappa}{2}(\mu-m)^2\}\propto\exp\{-\frac{1}{2}(n\tau+\kappa)\mu^2+(\tau n\bar{x}+\kappa m)\mu\}$: $A=n\tau+\kappa$, $B=\tau n\bar{x}+\kappa m$.

MCMC diagnostics: (1) Converged? (2) How correlated?

Trace plots: run several chains from **overdispersed** starts (more variable than the target). Good: chains overlap, stable, "hairy caterpillar". Separate or drifting: not converged. Slow wander, "woolly snake": high autocorrelation. **Burn-in:** discard early iterations affected by the start; read length off the trace.

BGR: M chains, each length $L+T$; keep last T. **B** = sd of the pooled MT draws (between). **W** = average of the M within-chain sds (within). $\hat{R}=B/W<1.05$: accept last T as stationary and pool. $\hat{R}\gg1$: not converged. $T\to\infty$: $\hat{R}\to1$. Exclude burn-in from $\hat{R}$.

Trace vs step size: tiny steps: smooth wander; huge steps: flat runs then jumps; good: fuzzy band. Accept near 1 and smooth: raise step. Low accept and plateaus: lower it. Gibbs accepts all, so use ACF, ESS, trace.

Not converged: run longer, better proposal. A pass is no proof: all starts may miss a region.

Autocorrelation: lag k: $\rho_k=\text{cor}(\theta^{(t)},\theta^{(t+k)})$; `acf(chain)`; `cor(chain[-n],chain[-1])` is $\hat{\rho}_1$. **Thinning:** keep every kth; simple, saves memory, throws away information. ESS never rises. `acf(x,plot=FALSE)$acf[2]` $=\hat{\rho}_1$ (entry 1 is lag 0); bars above the dashed band are real. `mean(diff(x)!=0)`: acceptance rate.

ESS: independent sample size with the same information: $n/(1+2\sum_i\rho_i)\approx n\frac{1-\rho_1}{1+\rho_1}$. Aim ESS >4000 (slides). E.g. $n=5000$, $\rho_1=.9$: ESS ≈263. Naive SE too small by $\sqrt{(1+\rho_1)/(1-\rho_1)}$. n for a target ESS: target×$\frac{1+\rho_1}{1-\rho_1}$.

Accuracy: $\text{Var}(\bar{h})\approx\frac{\sigma^2}{n}(1+2\sum\rho_i)$: iid SE is too small. MC s.e.: iid $s/\sqrt{N}$; proportion $\sqrt{\hat{p}(1-\hat{p})/N}$; chain $s/\sqrt{\text{ESS}}$. Posterior sd does not shrink with N. **Blocking:** MC error = sd(block means)$/\sqrt{Q}$; want $\le5\%$ of posterior sd. Same as ESS ≥400 (5% rule).

Maid Mast Codes PT4: MCMC

- `prob<-dtarget(prop)/dtarget(cur); if(runif(1)<prob) accept`: symmetric MH. `is.nan` guard. `proposal<-rnorm(2,chain[i,],sd)`: random walk.
- `acceptance<-1-mean(duplicated(chain[-(1:burnIn),]))`: after burn-in. `chain[-(1:burnIn),]` drops burn-in.
- `la1<-rgamma(1,a1+cfreq[M],b1+M)`: Gamma(shape, rate); `cfreq<-cumsum(freq)` gives $\sum_1^m y_i$.
- `M<-sample(1:n,1,prob=(la1/la2)^cfreq*exp((la2-la1)*(1:n)))`: discrete conditional.
- `rnorm(1,mu1+(x2-mu2)*s12/s2,sqrt(s1-s12^2/s2))`: Gibbs, sd not var.
- `cumsum(x)/1:n`: running mean, should flatten. `mean(th>0)`: posterior probability. `s[seq(1,n,by=k),]`: thin.
- `1/rgamma(1,a,b)`: InvGamma draw. `rnorm(1,m,sqrt(1/A))`: sd is the root of the variance.

VARIATIONAL INFERENCE

Variational inference: turns inference into optimisation: find $q^*\in D$ nearest the posterior. Faster for large data and complex models, at a cost in accuracy.

KL: $KL(q\|p(\theta|y))=\int q(\theta)[\log q(\theta)-\log p(\theta|y)]d\theta\ge0$. Not symmetric. Penalises q for mass where p is tiny, not for missing mass, so q sits inside p.

ELBO: $p(\theta|y)$ unknown, but $KL=-ELBO+\log p(y)$ with

$$ELBO(q)=-\int q[\log q-\log p(\theta)p(y|\theta)]d\theta$$

$=\mathbb{E}_q[\log p(\theta,y)]-\mathbb{E}_q[\log q]$. $\log p(y)$ is free of q, so **min KL = max ELBO**; $KL\ge0$ gives $\log p(y)\ge ELBO$. $\log p(y)=ELBO+KL$: the gap is the KL.

Mean field: $q(\theta)=\prod_j q_j(\theta_j)$: parameters independent under q. An assumption on the approximation, not on the model.

CAVI: update one factor at a time, others fixed: $q_j^*(\theta_j)\propto\exp\{\mathbb{E}_{-j}[\log p(\theta,y)]\}$. ELBO rises to a local optimum (a drop means a bug). Stop when ELBO change is small. Sensitive to initial values: run several, keep highest ELBO. It updates distributions; it draws no samples. Checks: ELBO never falls, stays $\le\log p(y)$.

Recipe: (1) $\log p(\theta,y)$ = log likelihoods + log priors. (2) Keep terms with θ_j. (3) Take $\mathbb{E}$ over the others (linear). (4) Collect powers, name the kernel. Same kernel as Gibbs, with terms in other parameters replaced by their expectations.

Use: $\mathbb{E}(x-\mu)^2=(x-\mathbb{E}\mu)^2+V\mu=\sigma^{2*}+\mu^{*2}-2x\mu^*+x^2$; $\mathbb{E}\mu^2=(\mathbb{E}\mu)^2+V\mu$; $\mathbb{E}\tau=a^*/b^*$; $\mathbb{E}\log\tau=\psi(a^*)-\log b^*$; $\mathbb{E}[1/\sigma^2]=a^*/b^*$. q_μ uses $\mathbb{E}[1/\sigma^2]$, not $1/\mathbb{E}\sigma^2$.

Normal-Gamma: $x_i\sim\text{N}(\mu,\frac{1}{\tau})$, $\mu|\tau\sim\text{N}(\mu_0,\frac{1}{\lambda_0\tau})$, $\tau\sim\text{Gamma}(a_0,b_0)$:
$q_\mu^*=\text{N}(\mu^*,\sigma^{2*})$: $\mu^*=\frac{\lambda_0\mu_0+n\bar{x}}{\lambda_0+n}$, $\sigma^{2*}=\frac{1}{(\lambda_0+n)\mathbb{E}\tau}$.
$q_\tau^*=\text{Gamma}(a^*,b^*)$: $a^*=a_0+\frac{n+1}{2}$, $b^*=b_0+\frac{1}{2}\sum_i\mathbb{E}(x_i-\mu)^2+\frac{\lambda_0}{2}\mathbb{E}(\mu-\mu_0)^2$.
Only σ^{2*} and b^* change between iterations. **Variance form** $\sigma^2\sim\text{InvGamma}(a_0,b_0)$, $\mu|\sigma^2\sim\text{N}(\mu_0,\sigma^2/\lambda_0)$: $q_{\sigma^2}^*$ is InvGamma with the same two parameters.

Independent priors $\mu\sim\text{N}(m,\frac{1}{\kappa})$, $\tau\sim\text{Gamma}(a,b)$: $q_\mu^*=\text{N}(\frac{\kappa m+n\bar{x}\mathbb{E}\tau}{\kappa+n\mathbb{E}\tau},\frac{1}{\kappa+n\mathbb{E}\tau})$; $q_\tau^*=\text{Gamma}(a+\frac{n}{2},\ b+\frac{1}{2}[\sum(x_i-\mathbb{E}\mu)^2+nV\mu])$. Shape $a+\frac{n}{2}$, not $\frac{n+1}{2}$; here both factors move each round. **Two sources,** y_j variance $d\sigma^2$: $\mathbb{E}Q=\sum(x_i-\mathbb{E}\mu)^2+nV\mu+\frac{1}{d}[\sum(y_j-\mathbb{E}\mu)^2+mV\mu]$, shape $a+\frac{n+m}{2}$.

Exact: $\mu|\tau,x\sim\text{N}(\frac{\lambda_0\mu_0+n\bar{x}}{\lambda_0+n},\frac{1}{(\lambda_0+n)\tau})$, $\tau|x\sim\text{Gamma}(a_0+\frac{n}{2},b_0+\frac{1}{2}\sum(x_i-\bar{x})^2+\frac{n\lambda_0(\bar{x}-\mu_0)^2}{2(\lambda_0+n)})$. $\text{Var}(\mu|x)=\frac{b_n}{(n+\lambda_0)(a_n-1)}$.

Bivariate Normal target: $\Lambda=\Sigma^{-1}$: $q_1^*=\text{N}(\mu_1-\frac{\Lambda_{12}}{\Lambda_{11}}(\mathbb{E}x_2-\mu_2),\frac{1}{\Lambda_{11}})$. $1/\Lambda_{11}=\sigma_1^2-\sigma_{12}^2/\sigma_2^2<\sigma_1^2$: the conditional variance. Means right, but VI underestimates variance and ignores correlation; worse as correlation rises. Error in the means shrinks by ρ^2 a round: strong correlation is slow too.

ELBO, Normal-Gamma: $\mathbb{E}[\log p(x,\mu,\tau)]-\mathbb{E}\log q_\mu-\mathbb{E}\log q_\tau$, with $\mathbb{E}\log q_\mu=-\frac{1}{2}\log\sigma^{2*}+c$, $\mathbb{E}\log q_\tau=\log b^*-\log\Gamma(a^*)+(a^*-1)\psi(a^*)-a^*$.

ELBO, bivariate target, up to a constant, $q_i=\text{N}(m_i,s_i^2)$: $-\frac{1}{2}[\Lambda_{11}(s_1^2+(m_1-\mu_1)^2)+\Lambda_{22}(s_2^2+(m_2-\mu_2)^2)+2\Lambda_{12}(m_1-\mu_1)(m_2-\mu_2)]+\frac{1}{2}\log s_1^2+\frac{1}{2}\log s_2^2$.

Bivariate prior (Gibbs row): $q_1^*=\text{N}(B/A,1/A)$ with μ_2 replaced by $\mathbb{E}\mu_2$.

Derivation, CAVI τ: $\log q_\tau^*=\mathbb{E}_\mu[\log p(x|\mu,\tau)+\log p(\mu|\tau)+\log p(\tau)]+c=(\frac{n}{2}+\frac{1}{2}+a_0-1)\log\tau-\tau[b_0+\frac{1}{2}\sum\mathbb{E}(x_i-\mu)^2+\frac{\lambda_0}{2}\mathbb{E}(\mu-\mu_0)^2]+c$: power of $\log\tau$ is a^*-1, coefficient of $-\tau$ is b^*.

MCMC vs VI: MCMC simulates the density, is heavier, asymptotically exact. VI approximates it, is faster, suits large data and model exploration, no guarantee, underestimates variance.

EXAM CHECKS

- Define P_{ik}; state which θ^0.
- State H_0, statistic, df, decision, sentence in context.
- CI on η first, transform last.
- $\hat{p}$, $\hat{\mu}$ on response scale.
- p includes intercept.
- $\hat{\phi}$: big model for F, own model for SE.
- Rate vs scale; variance vs sd in N.
- Name distribution and every parameter.
- 4 dp inside, round once.
- N(mean, variance), Gamma(shape, rate), InvGamma(shape, scale).